

Contact

yunchuangao@gmail.com

www.linkedin.com/in/yunchuan-gao-1075197 (LinkedIn)

Top Skills

Statistics

Machine Learning

Quantitative Analytics

Certifications

M3 - Rules & Regulations for Fund Management

M5 - Rules And Regulations For Financial Advisory Services

M6 - Securities Products & Analysis

Publications

Multi-category Support Vector Machines

Fitting Mixture Distributions Using Generalized Lambda Distributions and Comparison with Normal Mixtures

Fitting Mixture Distributions Using a Mixture of Generalized Lambda Distributions with Computer Code

Yunchuan Gao

Algorithmic Trading, Asset Management, Machine Learning Expert
Toronto, Ontario, Canada

Summary

15+ years experience in algorithmic trading strategy and asset management.

15+ years experience in machine learning applications in hedge fund systematic trading, asset allocation and health & safety.

Expert in machine learning, mathematical optimization, numerical analysis, statistical analysis and portfolio construction.

Ph.D. in Applied Mathematics, M.S. in Computer Science, M.S. in Statistics, B.S. in Applied Mathematics.

Experience

CIBC

Consultant

June 2023 - Present (3 years 3 months)

Develop machine learning models for direct investing.

AI Numeric

Founder

July 2020 - Present (6 years 2 months)

Provide asset allocation solution for asset management companies.

Develop quant equity trading models for hedge funds.

Provide machine learning training and education.

Eastspring Investments

Quant Portfolio Manager

September 2021 - June 2023 (1 year 10 months)

Manage quant portfolio of AUM USD 1 Billion.

Develop change point and machine learning alpha signals.

Research and develop NLP based sentiment alpha signals using S&P data.

CPP Investment Board

Portfolio Manager

November 2011 - May 2020 (8 years 7 months)

Toronto, Canada

Lead portfolio construction team to deliver quantitative strategies to apply in multi-asset class portfolio rebalancing.

Conduct research in machine learning applications in online portfolio optimization.

Develop machine learning clustering model to manage portfolio management constraints.

Develop and manage quantitative portfolio rebalancing strategy for long/short portfolio.

Proficient in Python, MATLAB.

The QWeMA Group

Senior Researcher

May 2011 - November 2011 (7 months)

Toronto, Canada

Develop quantitative Robo-Advisor product.

Develop Retirement Wealth Management System using optimal control theory; deliver product to insurance companies in U.S.

Proficient in C++, MATLAB.

Mapleridge Capital Corporation

Senior Quantitative Strategist

September 2006 - May 2011 (4 years 9 months)

Toronto, Canada

Manage USD long-short portfolios in global futures market.

Lead research and development team.

Build algorithmic high frequency futures trading strategies; trading signals generated by Support Vector Machine forecast model; apply Bayesian based online change point detection model.

Develop and manage clusters of portfolio rebalancing strategies: dynamic rebalancing; machine learning online rebalancing; intra-day rebalancing strategies.

Conduct research and build prototype of ultra-high frequency trading strategies using level 2 tick data.

Proficient in C++, MATLAB.

Orchard Asset Management

Stock Trading Strategist

January 2006 - July 2006 (7 months)

Toronto, Canada

Develop U.S. stock market pair trading strategy.

Develop automated trading reconciliation system.

Proficient in C++.

Education

Syracuse University

Doctor of Philosophy - PhD, Applied Mathematics

Syracuse University

Master of Science, Computer Science

Syracuse University

Master of Science, Mathematics

Indiana University of Pennsylvania

Master of Science, Applied Statistics

Tongji University

B.S., Applied Mathematics